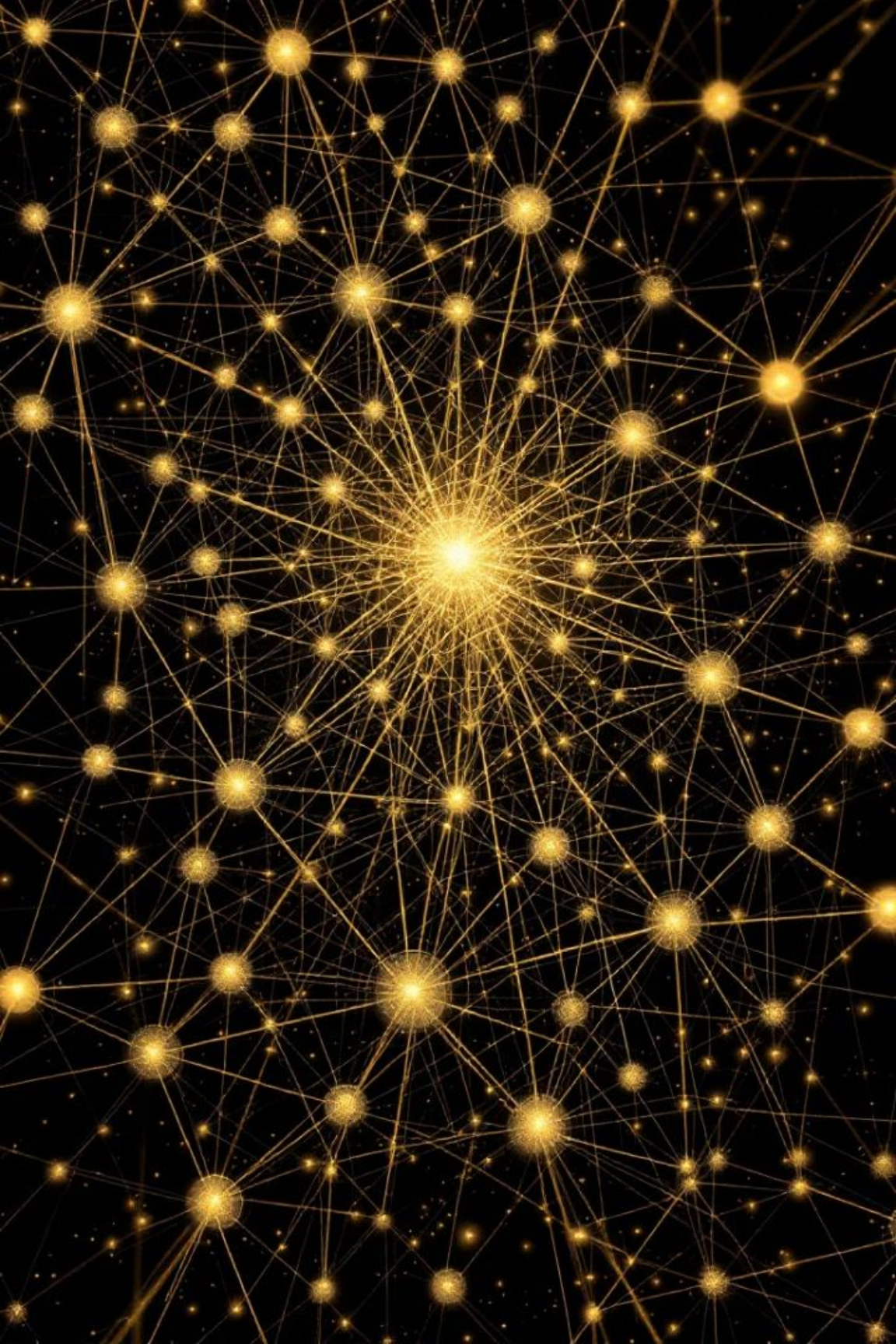




Associative Learning: Understanding How We Connect Connect the Dots

Associative learning is a fundamental process. It explains how we connect events and stimuli. This forms the basis of our understanding of the world.

 by PRANJALI BAHALKAR



Associative Learning in Artificial Neural Networks

This presentation explores associative learning, a powerful technique used in artificial neural networks (ANNs) to model complex relationships and solve challenging problems.

Associative learning is a style of learning that happens when two unrelated elements (for example, objects, sights, sounds, ideas, and/or behaviours) become connected in our brains through a process known as conditioning.

What is Associative Learning?

Associative learning is a type of machine learning where an ANN learns to associate different inputs and outputs, forming complex relationships.

Think of it as connecting the dots - the ANN identifies patterns and links between different pieces of information, like identifying a specific object based on its features.

Conti..



Understanding Associative Learning in ANN

Exploring the mechanisms and applications in ANN



Overview of Associative Learning

This concept involves learning associations between stimuli and responses in ANNs.



Definitions of Associative Learning

It refers to a learning process where a connection is made between two stimuli through experience.



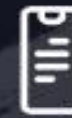
Mechanisms in Associative Learning

Explores the algorithms and processes that enable ANNs to learn from associations.



Applications in Various Fields

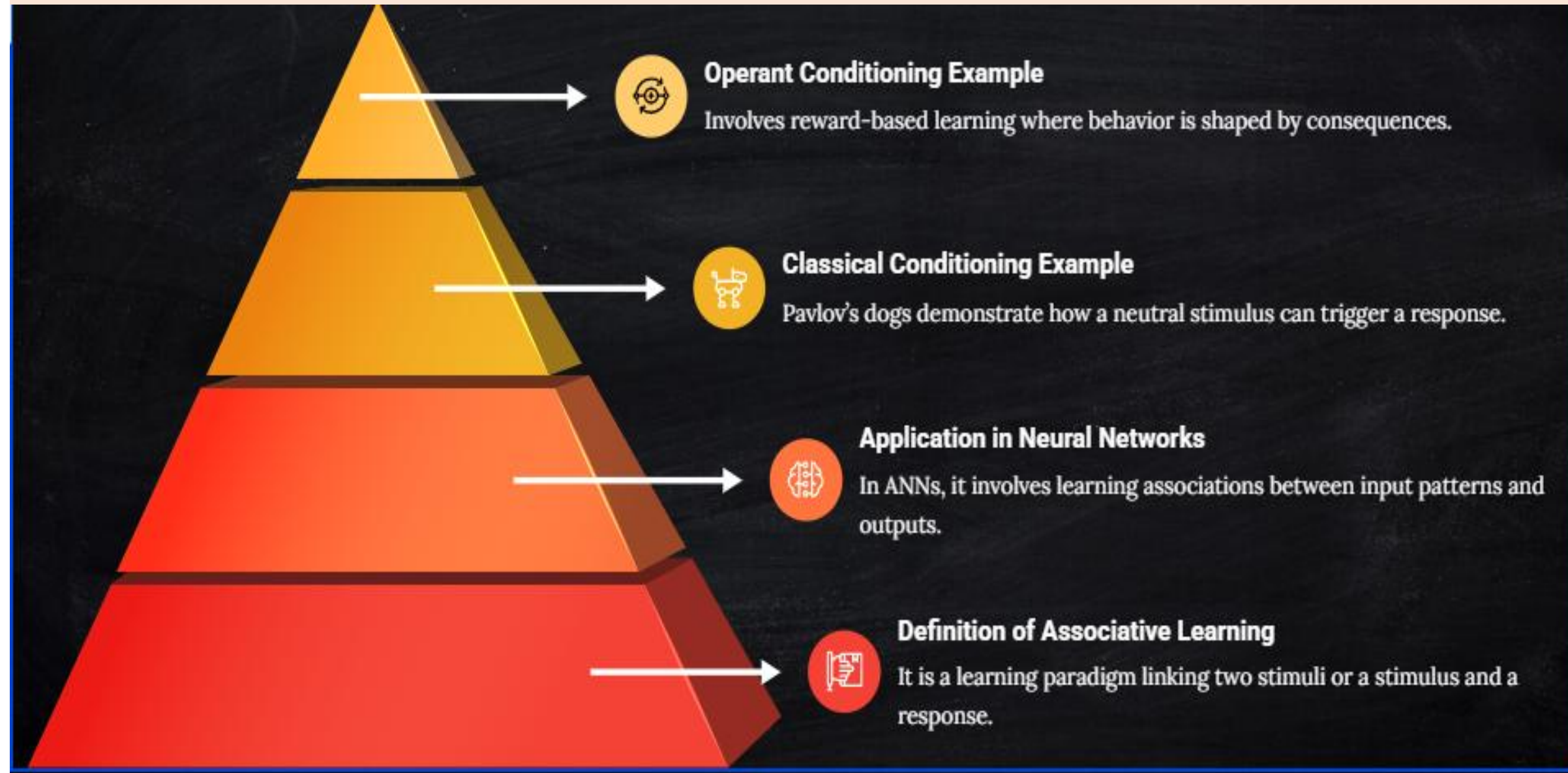
Associative learning is applied in areas such as pattern recognition, speech processing, and robotics.



Future Directions in ANNs

Research is focusing on enhancing associative learning to improve ANN efficiency and adaptability.

Understanding Associative Learning



Examples of associative learning include:

- If someone puts their hand on a hot stove and hurts themselves, they may learn to associate hot stoves with pain, and have therefore been conditioned not to put their hands on them.
- If someone eats a particular food, then develops a headache soon afterwards, they may learn to associate that food with headaches (even if the food didn't cause the headache), and not want to eat it again.
- Every time a child cleans their room, their parent or carer gives them a treat. The child starts associating cleaning their room with treats, making them more inclined to clean their room more frequently.

The title of for:

Pavlov's



Classical Conditioning: Pavlov's Dogs

1

Unconditioned Stimulus

Food naturally triggers salivation.

2

Neutral Stimulus

Bell initially causes no response.

3

Conditioned Stimulus

Bell, after association, triggers salivation.

4

Conditioned Response

Salivation to the bell alone.

1. Before Conditioning:

- **Unconditioned Stimulus (US):** Food
- **Unconditioned Response (UR):** Salivation (Dogs naturally salivate when they see/smell food)
- **Neutral Stimulus (NS):** Bell (Before conditioning, the bell does not cause salivation)

2. During Conditioning (Acquisition Stage):

- Pavlov rang a bell (NS) just **before** presenting food (US) to the dog.
- The dog **learned to associate** the bell with food through repeated pairings.

3. After Conditioning:

- The **bell alone** (now the **Conditioned Stimulus (CS)**) caused the dog to salivate, even when no food was presented.
- The **salivation response** to the bell became a **Conditioned Response (CR)**.

Operant Conditioning:

- When a kitten is misbehaving, its mother will flick its ears. The kitten eventually learns to associate misbehaving with ear flicking (which is painful to them), so it stops.



Conti..

- Associative learning is something that all humans and animals do naturally. By linking elements together and making a web of different connections, we build up our memories and deepen our understanding of the world around us. If we did not do this, we would not be able to recall even the most basic of things, such as how to get to the local shops, or that we do not like certain foods.
- As well as being something that humans and animals do naturally, associative learning is also utilised by those who teach. Through using associative learning techniques, teachers are better able to manage their classrooms, while parents and carers are better able to encourage their children to behave well and responsibly.



Real-World Examples: Classical Conditioning



Marketing

Associate products with positive emotions. This increases consumer appeal.



Taste Aversion

Avoid foods that caused illness. This is a survival mechanism.



Emotional Response

Certain songs trigger specific memories. These evoke strong emotions.

Operant Conditioning in Action



Training Animals

Use rewards to teach tricks. This reinforces desired actions.



Shaping Human Habits

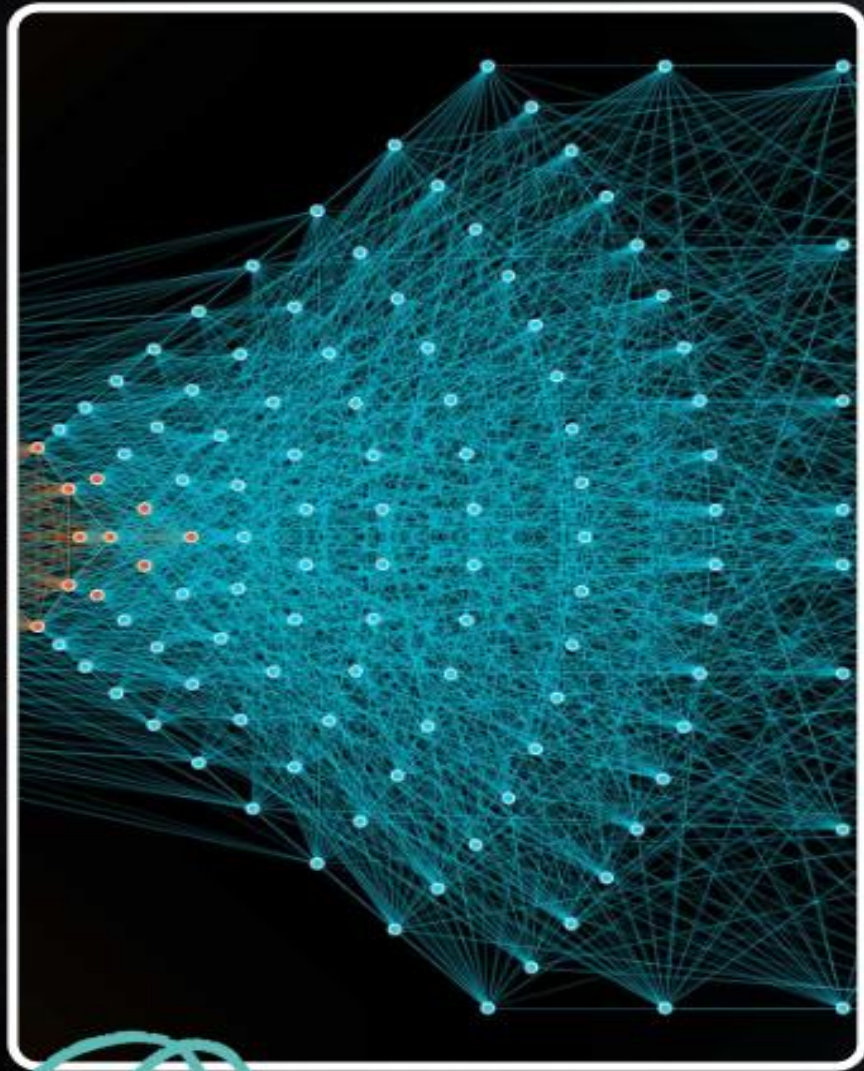
Implement reward systems. Achieve personal and professional goals.



Child Education

Encourage good behavior with praise.
Discourage bad behavior with consequences.

The Importance of Associative Learning in ANN



The Importance of Associative Learning in ANN

Understanding its Role in Enhancing AI Performance

- 01 Generalization from Learned Data**
Associative learning enables networks to apply knowledge to new, unseen data effectively.
- 02 Pattern Recognition in Complex Datasets**
Facilitates the identification of patterns and relationships, crucial for AI applications.
- 03 Enhanced Adaptability of Neural Networks**
Improves performance across tasks like classification, regression, and clustering.

Types-

Understanding Associative Learning

Key Concepts in Learning Theory

01 Classical Conditioning

A learning process where a neutral stimulus becomes associated with an unconditioned stimulus, leading to a conditioned response.

02 Operant Conditioning

Learning influenced by the consequences of behavior, incorporating rewards and punishments to shape actions.

03 Hebbian Learning

A principle stating that synaptic connections strengthen when neurons are activated simultaneously, enhancing learning.



Types of Associative Learning

- There are **two types** of associative learning:
- classical conditioning
- operant or instrumental conditioning.

Classical conditioning

- **Classical conditioning** is when one neutral element (called the conditioned stimulus) becomes associated with a different element (the unconditioned stimulus) that already generates the desired response (the unconditioned response), until the neutral element stops being neutral, and instead causes the same response (the conditioned response).
- One very famous example of classical conditioning was conducted by Russian scientist Ivan Pavlov, where he conditioned dogs to salivate whenever they heard a bell.

Conti..

- To do this, he would ring a bell (the conditioned stimulus), then give the dogs food (the unconditioned stimulus) very soon afterwards. The food caused the dogs to salivate (the unconditioned response), but after some time doing this bell ringing/food giving routine, Pavlov noticed that the dogs would salivate immediately after the bell was rung - even before they saw any food. Pavlov then tried ringing the bell without then producing the food, and the dogs still salivated once the bell was rung (the conditioned response). At this point, Pavlov concluded the dogs had been conditioned to salivate at the sound of the bell.



Operant conditioning

- **Operant conditioning** is quite similar to classical conditioning, but instead of conditioning an unconscious response, operant conditioning is about conditioning an individual's conscious behaviour, based on what the understood consequences of that behaviour will be.
- One well-known example of operant conditioning comes from the psychologist B.F. Skinner, who conditioned rats to perform behaviours using positive and negative reinforcements.

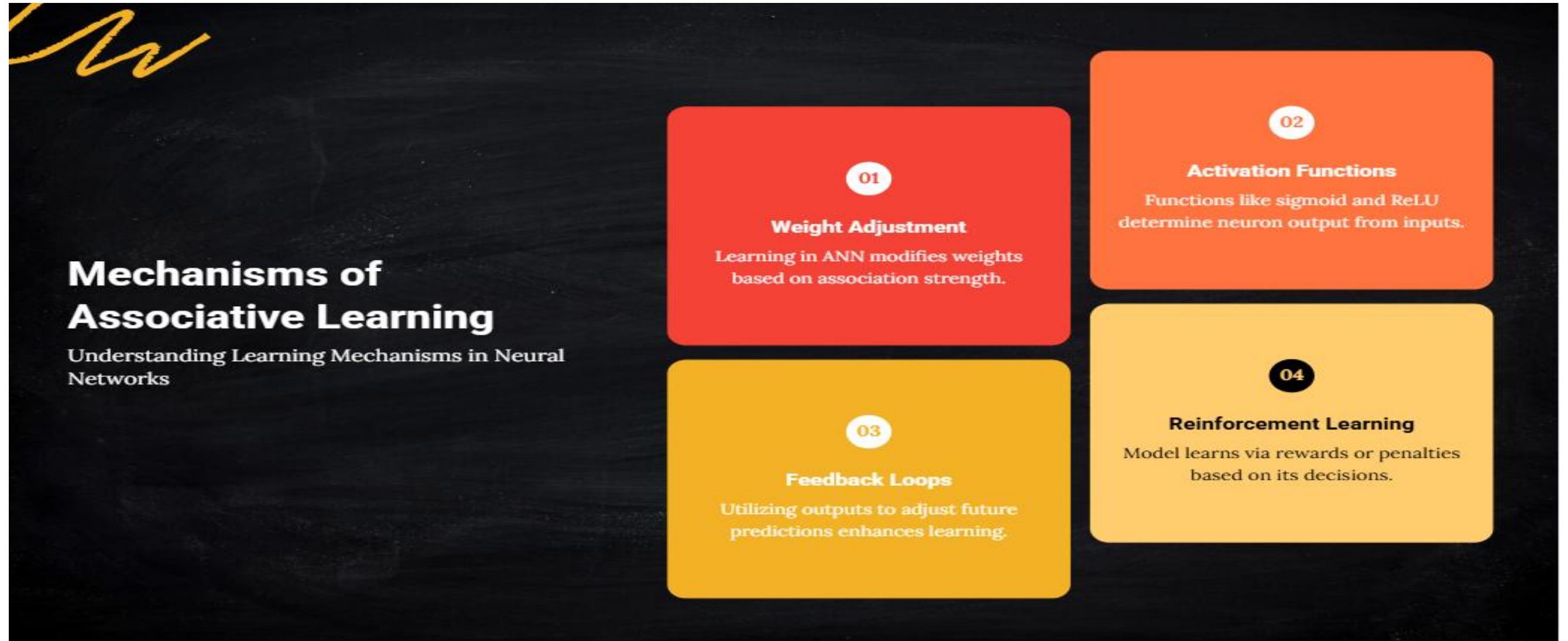
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- In his first experiment, he put a hungry rat into a specially designed Skinner box, which was equipped with a lever mechanism that would dispel food when pressed. After spending some time exploring the box, the rat pressed the lever and the food appeared. Skinner repeated this experiment with the same rat several times, until eventually the first thing the rat would do upon entering the box would be to press the lever. In this instance, a positive reinforcement (the reward of food) had been used to condition the rat to push the lever.

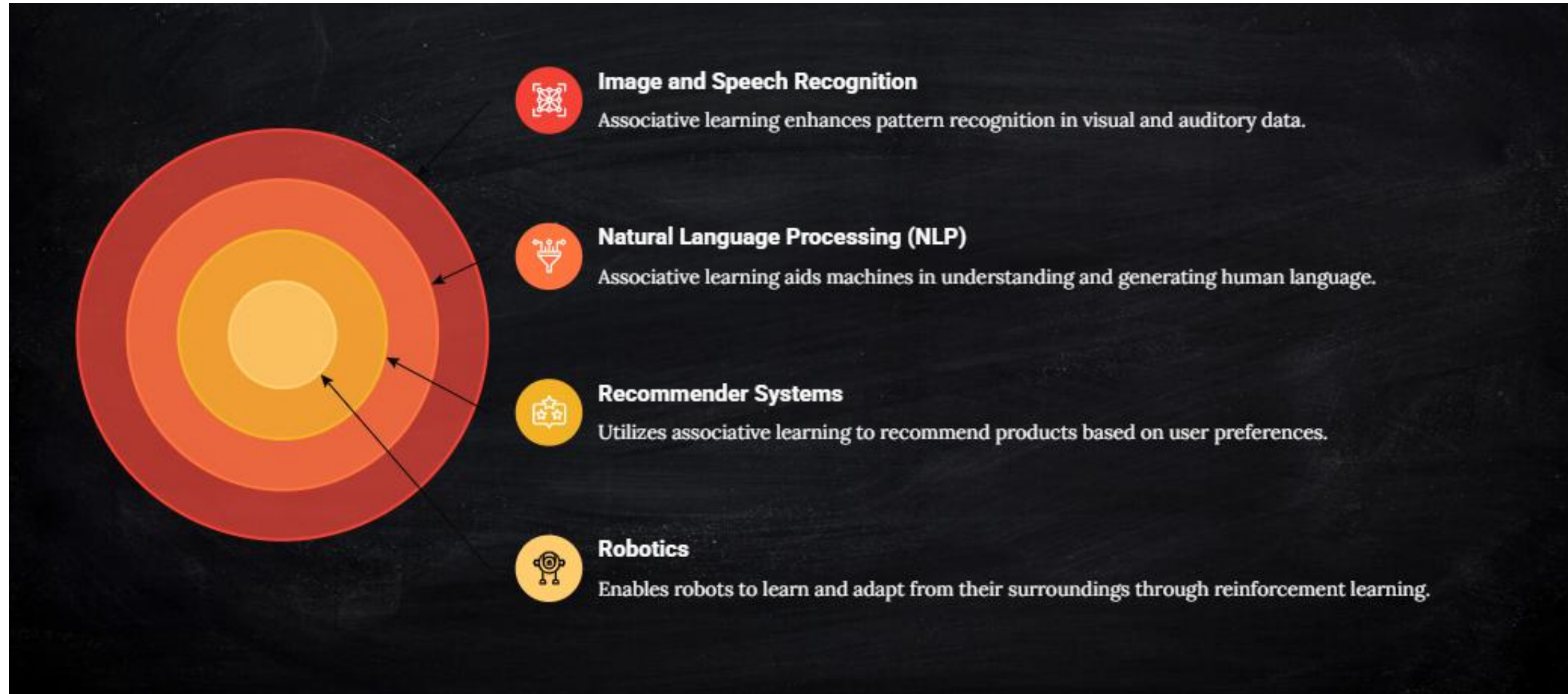


- In his second experiment, Skinner put a different rat in a Skinner box that had an uncomfortable electric current running through it. The rat, having experienced the discomfort, moved around the box and accidentally knocked the lever, which immediately stopped the current. Again, Skinner repeated the experiment several times, until the rat knew to press the lever immediately to avoid discomfort. Here, a negative reinforcement (the possibility of discomfort from the current) was used to condition the rat to pull the lever.

Mechanisms of Associative Learning-

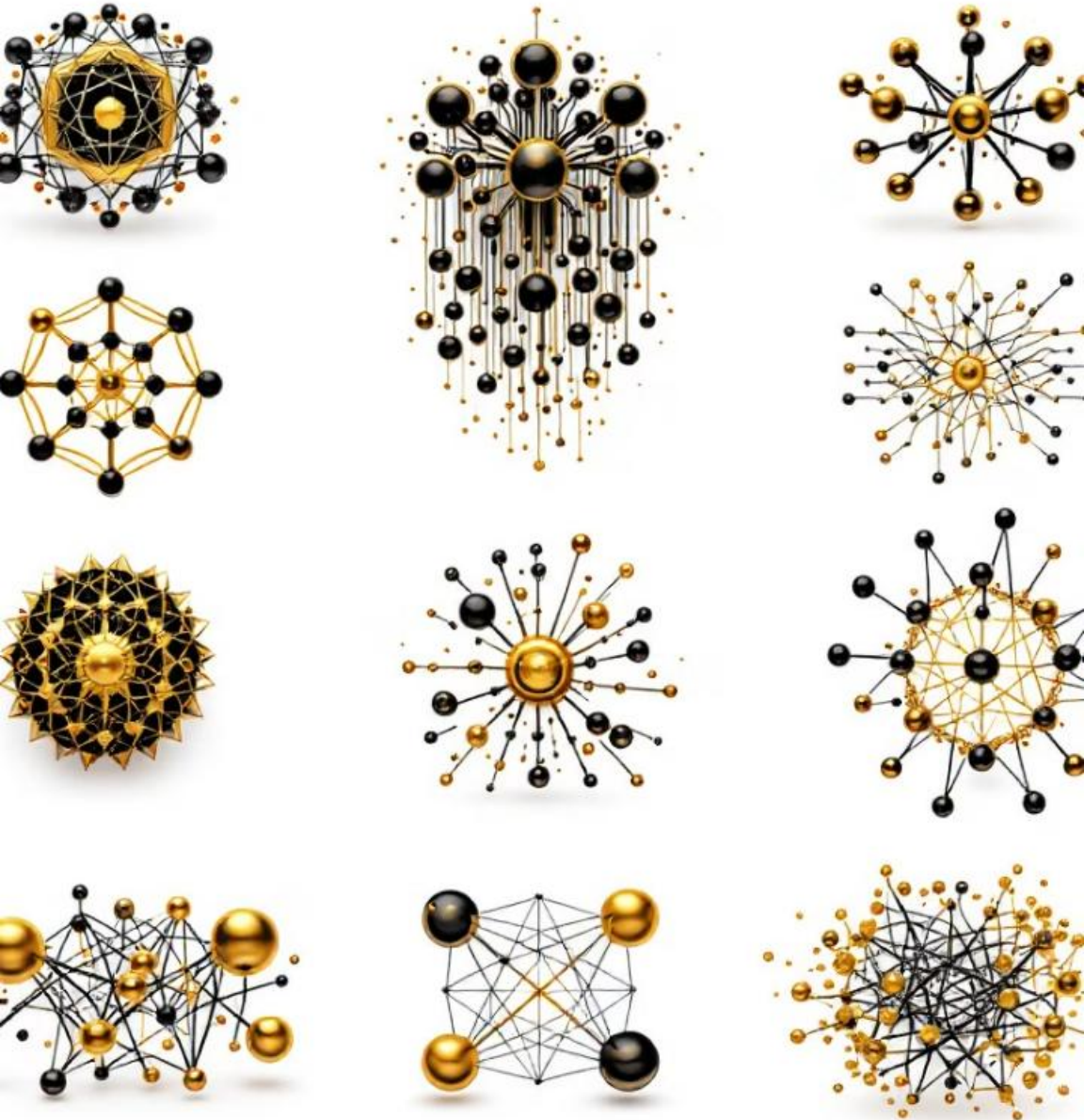


Applications of Associative Learning





Implementing Associative Learning in ANNs: Key Architectures



Hopfield Networks

Designed for storing and retrieving patterns based on partial input, with applications in image recognition and memory.

Kohonen Self-Organizing Maps

A type of unsupervised learning where neurons organize themselves based on similarities in input data, ideal for clustering and data visualization.

Recurrent Neural Networks (RNNs)

Designed for sequential data, RNNs have internal memory allowing them to learn from past inputs, perfect for language modeling and time series analysis.

Hopfield network

- The Hopfield networks are categorized into two categories. These are:
- **Discrete Networks**
- These networks give any of the two discrete outputs. Based on the output received, further two types:
 - 1.Binary:** In this type, the output is either 0 or 1.
 - 2.Bipolar:** In bipolar networks, the output is either -1 (When output < 0) or 1 (When output > 0)

$$\text{State} = \begin{cases} -1; & \text{Output} < 0 \\ 1; & \text{Output} > 0 \end{cases}$$

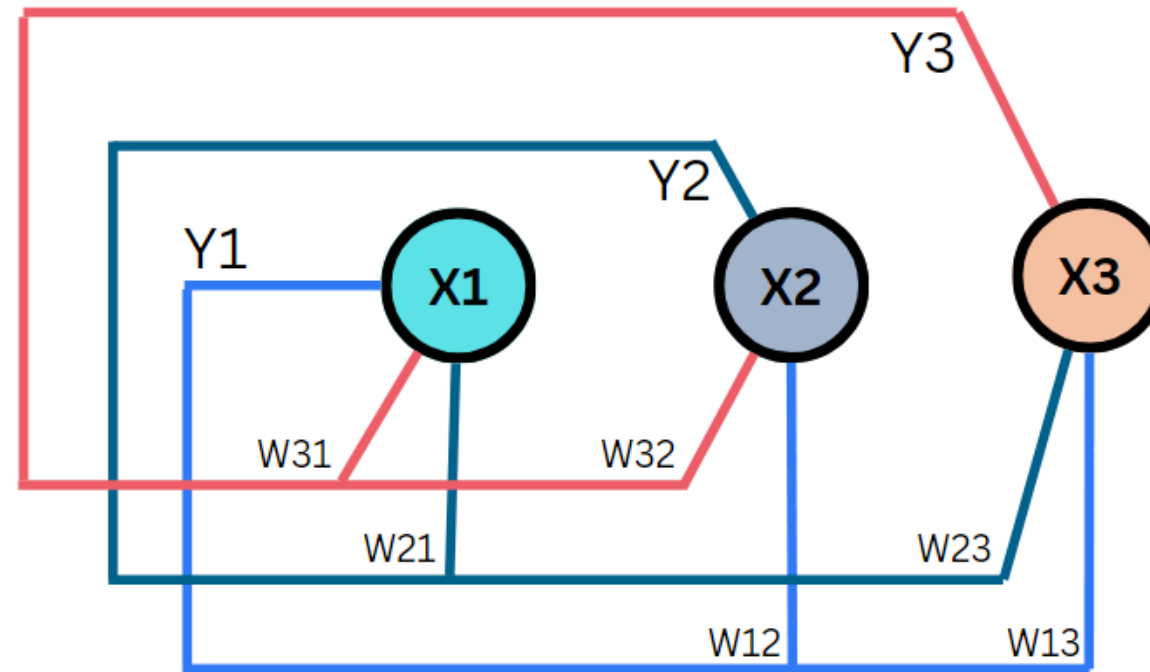
Continuous Networks

- Instead of receiving binary or bipolar output, the output value lies between 0 and 1.

The Architecture of Hopfield Network

- The architecture of the Hopfield network consists of the following elements:
- Individual nodes preserve their states until required an update.
- The node to be updated is selected randomly.
- Each node is connected to all other nodes except itself.
- The state of each node is either 0/1 or 1/-1.
- The Hopfield network structure is symmetric, i.e., $W_{ij} = W_{ji}$ for all i 's and j 's.

The representation of the **sample architecture of the Hopfield network** having three nodes is as follows:



In the above diagram, each symbol represents:

x_1, x_2, x_3 - represents the input.

y_1, y_2, y_3 - represents output obtained from each node.

W_{ij} - represents the weight associated with the connection from i to j .

Training Model with Hopfield Network

- Training a Hopfield network refers to lowering of energy of each state. The training model consists of a training and testing algorithm. Let's discuss each of them one by one.

Training Algorithm

- It is based on the Hebbian principle, which Donald Hebb gave in 1949. In training, algorithm weights are updated using a specific rule. The rule is different for binary and bipolar. For storing a set of 'N' input patterns [$X(n)$ where $n = 1$ to N],

Rule followed is as -

- Binary Input:

$$W_{ij} = \sum_{n=1}^N (2X_i(n) - 1) (2X_j(n) - 1)$$

- Bipolar Input:

$$W_{ij} = \sum_{n=1}^N (X_i(n) X_j(n))$$

Testing Algorithm

The testing algorithm involves several steps.

These steps are:

- Initialize the weights using the above training algorithm rules.
- Follow steps 3 to 7 for each input vector ' X_i '.
- Assign the value of external input vector ' X_i ' to initial activators ' Y_i ', for all $i = 1$ to N .
- Follow steps 5 to 7 for each initial activator ' Y_i '.
- Calculate the network's net input ' Y_{in} ' using the below rule:

Conti..

$$Y_{ij} = X_i + \sum_{i=1}^N Y_i W_{ij}$$

- Based on the calculated net input and threshold value, calculate the output. There are three different conditions for the output. These are:

$$Y_i = \begin{cases} 1, & \text{if } Y_{in} < \theta \\ Y_i, & \text{if } Y_{in} = \theta \\ 0, & \text{if } Y_{in} < \theta \end{cases}$$

- Broadcast the obtained output 'Y_i' to all other units. And update the activation vector.
- Test the network.

Error Performance in Hopfield networks

1. How Does a Hopfield Network Work?

- ✓ It remembers patterns like our brain remembers faces.
- ✓ When given an incomplete or noisy version of a pattern, it tries to fix it and recall the correct one.
- ✓ It does this by updating neuron connections until it reaches a stable pattern.

2. Why Do Errors Happen?

- Errors in Hopfield networks occur due to:

- ✓ **Too Many Patterns** – If we store too many patterns, the network gets confused.

(Max safe limit = $0.15 \times \text{number of neurons}$)

- ✓ **Similar Patterns** – If two patterns look alike, the network might mix them up.

- ✓ **Noise & Distortion** – If the input is too different from the stored pattern, the network may retrieve the wrong one.

- ✓ **False Memories (Spurious States)** – The network sometimes creates new patterns that weren't stored.

Example:

- Suppose we have a Hopfield network with **10 neurons**.
- The safe limit for storing patterns is **$0.15 \times 10 = 1.5$ patterns** (rounded to 1 or 2 patterns).
- If we try storing **5 patterns**, the network starts to confuse them, leading to errors.

Impact:

- The network recalls a **blended or incorrect** pattern instead of the original.
- The **retrieval accuracy drops** significantly as the number of stored patterns increases.

Why Does It Happen?

- If two stored patterns look **too similar**, their weight matrices overlap.
- When a noisy or incomplete pattern is given as input, the network may mistakenly recall the wrong one.

Example:

- Suppose we store two patterns:
 - $P1 = (1, -1, 1, -1)$
 - $P2 = (1, -1, -1, 1)$
- These patterns differ by only **two positions**.
- If we give the input $(1, -1, ?, ?)$, the network might **confuse P1 and P2**, leading to errors.

How to Reduce Errors?

- **Store fewer patterns** to avoid interference (stay within the $0.15N$ limit).
- **Modify learning rules** (e.g., Stochastic Hopfield Networks) to handle false memories.

Hopfield networks work well for **associative memory**, but their performance drops if **too many patterns are stored, inputs are too noisy, or patterns are too similar**. By optimizing how we store and retrieve patterns, we can improve accuracy and reduce errors.

Example scenario:

- Imagine a Hopfield network trained to recognize simple handwritten digits (like "0", "1", "2") as stored patterns.

Correct recall:

- If the network is presented with a slightly noisy version of a "1" (e.g., with a few pixels flipped), it should ideally converge to the clean "1" pattern after several iterations.

Error occurrence:

- **High noise:** If the input is heavily corrupted with noise, the network might misinterpret the pattern and converge to a different stored digit (e.g., "7").
- **Overlapping patterns:** If the network is also trained to recognize "7" and the input is slightly distorted, it might mistakenly identify it as a "7" due to the similarity between the two patterns.

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Measuring error performance:

Hamming distance:

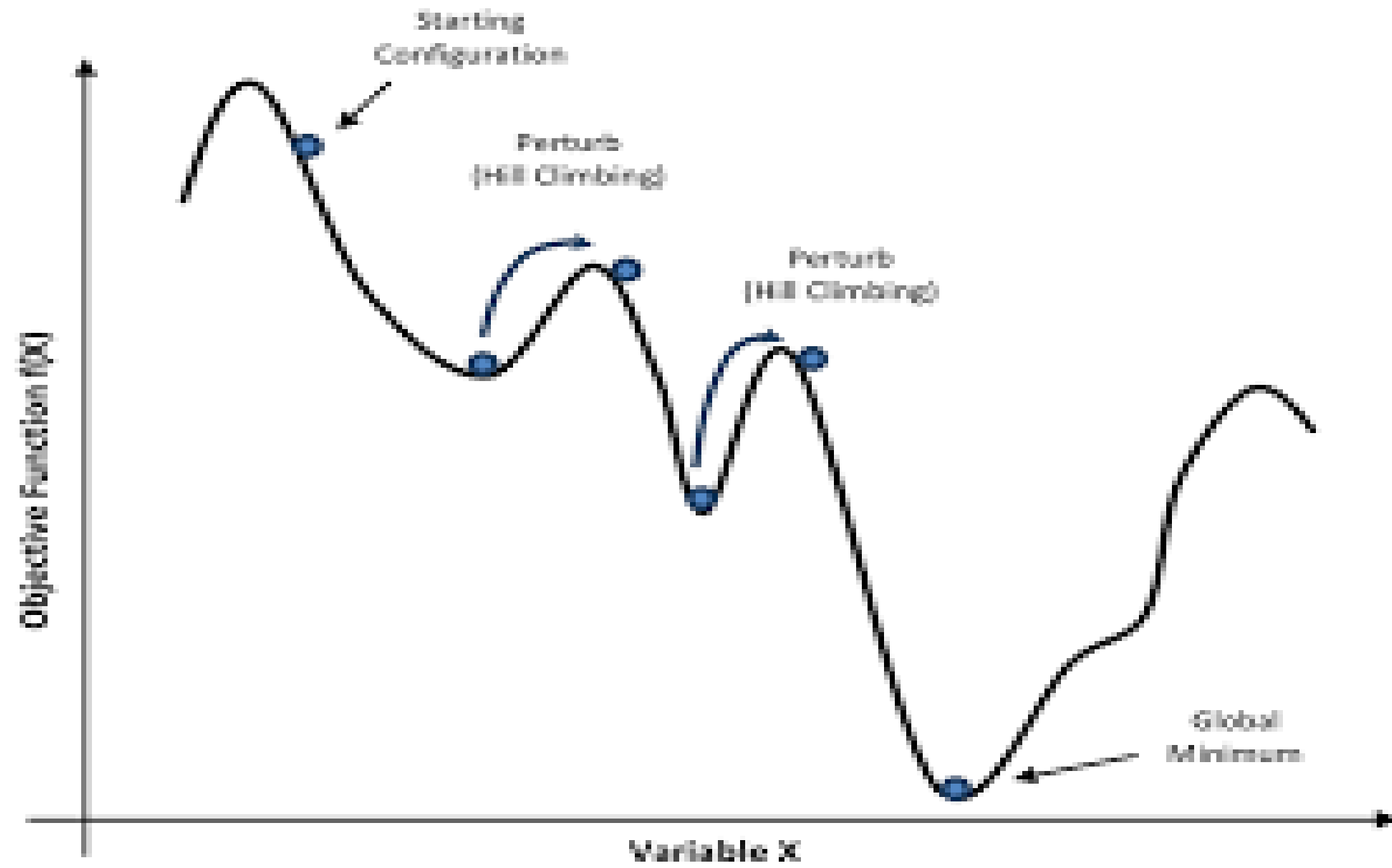
- A common metric to quantify errors is the Hamming distance between the network's output and the target pattern, which counts the number of differing bits.

Recall rate:

- The percentage of correctly recalled patterns out of a set of test inputs is another way to evaluate error performance.

Simulated Annealing-

- Simulated Annealing (SA) is used in **Artificial Neural Networks (ANNs)** to optimize weights, reduce errors, and avoid local minima during training. Instead of using traditional methods like **Gradient Descent**, SA provides a way to escape poor solutions and find better ones.
- **Why Use Simulated Annealing in ANNs?**
- **✓ Avoids Local Minima** – Unlike gradient-based methods (e.g., Backpropagation), SA can escape local minima and find better solutions.
- **✓ Good for Non-Differentiable Functions** – Works even when derivatives don't exist or are hard to compute.
- **✓ Useful for Hyperparameter Tuning** – Helps find the best learning rate, number of neurons, or network structure.



What is Simulated Annealing?

- **Definition:** Simulated annealing is an optimization method inspired by metallurgy, where metals are heated and slowly cooled to remove defects and achieve a stable, optimal state.
- **Purpose:** It helps find a good solution to complex optimization problems by exploring different possibilities and avoiding getting stuck in suboptimal solutions (local optima).

Key Concepts

- Optimization:** The process of finding the best possible solution among many options.
- Global vs. Local Optima:**
 - Global Optima:* The absolute best solution across the entire solution space.
 - Local Optima:* The best solution within a limited region but not necessarily the best overall.
- Annealing:** The gradual reduction of temperature to stabilize the system, analogous to fine-tuning the solution in simulated annealing.

How Simulated Annealing Works in ANNs?

Step 1: Initialize the Network

- Randomly set the ANN weights and biases.

Step 2: Compute the Error (Energy Function)

- Use a loss function (e.g., Mean Squared Error, Cross-Entropy) to measure how well the network performs.

Step 3: Make Small Changes (Neighbor Solutions)

- Slightly adjust weights and biases to explore new solutions.

Conti..

Step 4: Accept or Reject the New Solution

- **If the new solution is better**, accept it.
- **If worse, accept it with some probability** (which decreases over time).
- This helps escape poor local minima.

Step 5: Reduce the Temperature

- Slowly decrease the probability of accepting bad solutions as the process continues.

Step 6: Stop When Converged

- The network settles into an optimized weight configuration.

Explanation-

- Initial Solution:**

Start with any random or basic solution to the problem.

- Evaluation:**

Measure the quality of the solution using a cost or objective function.

- Generate a Neighbor Solution:**

Make a small random change to the current solution to explore new possibilities.

Conti..

- **Acceptance Criteria:**

If the new solution is better: Accept it.

If the new solution is worse: Accept it with a probability that decreases as the system “cools.” This helps avoid getting stuck in local optima.

Cooling Schedule:

Gradually reduce the temperature parameter to shift from exploration to exploitation.

Repeat:

Continue this process until the system is “frozen,” meaning no further changes improve the solution significantly.

Why Accept Worse Solutions?

- Accepting worse solutions occasionally allows the algorithm to jump out of local optima and explore a broader solution space.

- **High Temperature:**

During the early stages, the algorithm is more likely to accept worse solutions, promoting exploration.

- **Low Temperature:**

As the temperature decreases, the acceptance of worse solutions is reduced, focusing on refining the best-found solutions.

Example: Traveling Salesman Problem (TSP) Problem:

- A salesman must visit N cities and return to the starting city while minimizing travel distance.
- How SA Helps?
- Start with a random tour of cities.
- Swap two cities and check if the new route is shorter.
- If shorter, accept it. If longer, accept it with some probability.
- Gradually reduce acceptance of bad swaps as the "temperature" decreases.
- Eventually, find a near-optimal or optimal route.

Real-Time Example-

- **Traveling Salesman Problem:** Simulated annealing can find a near-optimal path for visiting multiple cities with the shortest total distance.
- **Machine Learning:** It helps in hyperparameter tuning to enhance model performance.
- **Manufacturing:** Optimizes scheduling and resource allocation for improved efficiency.

Conti..

Where is Simulated Annealing Used?

- ✓ Machine Learning & AI – Fine-tuning models.
- ✓ Circuit Design – Optimizing layouts for lower power usage.
- ✓ Scheduling Problems – Assigning tasks efficiently.
- ✓ Route Optimization – Finding shortest delivery routes.

Advantages & Disadvantages-

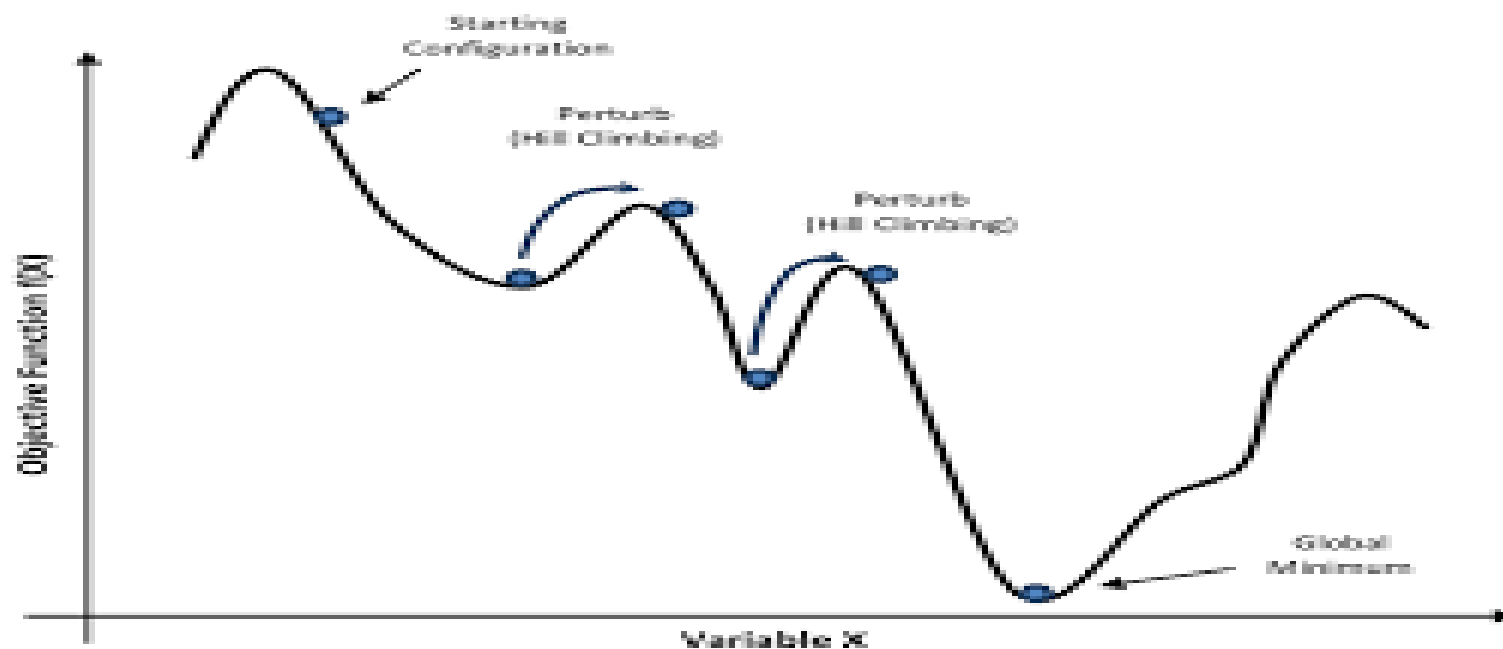
- **Advantages:**

- Simple and adaptable to various types of problems.
- Effective for large and complex optimization scenarios.

- **Disadvantages:**

- The convergence to the optimal solution can be slow.
- The performance highly depends on the cooling schedule chosen.

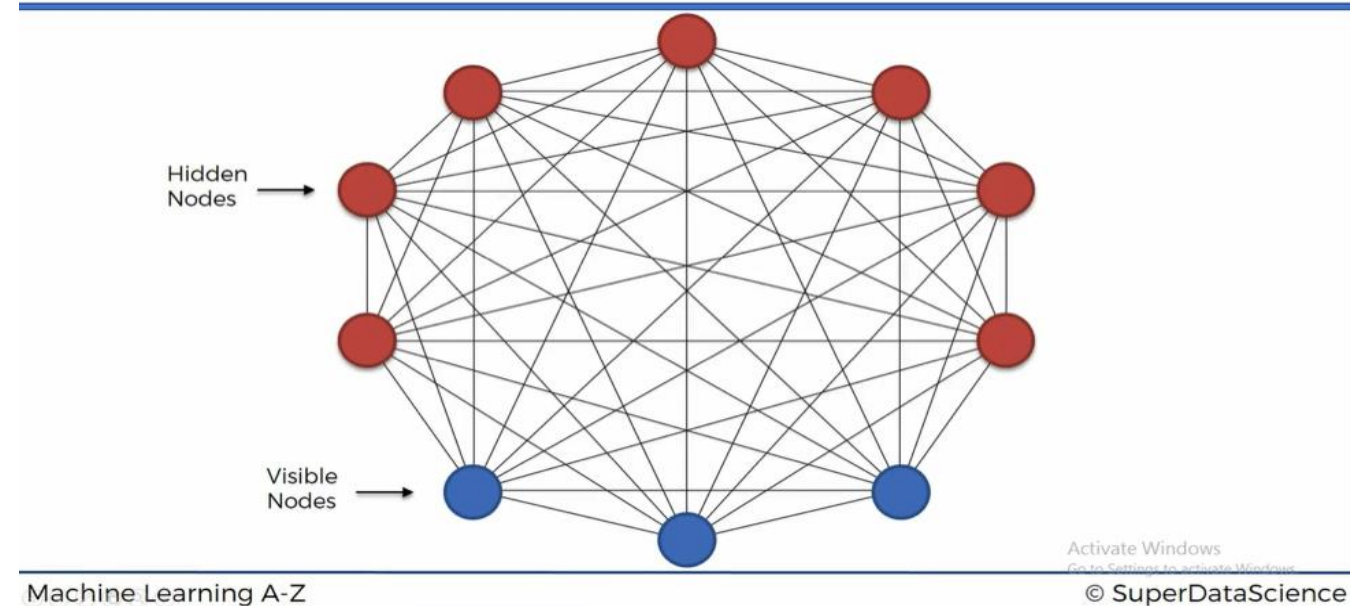
- Simulated annealing is a robust optimization technique that strikes a balance between exploring new solutions and exploiting the best ones found. Particularly useful for problems where the solution landscape is complex with many local optima.
- Simulated annealing (SA) is a computer science technique that finds the best solution to an optimization problem. It's inspired by the process of heating and then cooling a material to reduce defects.



Boltzmann machine and Boltzmann learning

- A **Boltzmann Machine (BM)** is a type of stochastic recurrent neural network with a structure similar to a Hopfield network but with stochastic (random) neuron behavior. It is named after the Boltzmann distribution in statistical mechanics. BMs are primarily used for learning complex probability distributions over binary vectors and for solving optimization problems.

Boltzmann Machines



Types-

- Boltzmann machines have two main types of units, like players in a team - some in the spotlight and some behind the scenes.

- Visible units-

The visible units are like the forwards scoring goals; they interact directly with the outside world.

- The hidden units-

The hidden units are like the strategists planning the game; they work in the background. These units work together to make the machine "learn" patterns and make decisions.

Key Components of a Boltzmann Machine:

- **Neurons (Nodes):** These can be either visible (representing observed data) or hidden (capturing dependencies and patterns).
- **Weights:** Symmetrical weights between neurons determine the network's energy.
- **Energy Function:** Defines the probability of the network being in a particular state. Lower energy states are more probable.
- **Stochastic Neuron Activation:** The neurons' states are updated probabilistically according to the Boltzmann distribution.

Boltzmann Learning:

- Boltzmann learning is the process of training a Boltzmann Machine. The goal is to adjust the weights of the network to minimize the difference between the data distribution and the model distribution.
- Boltzmann learning is a stochastic learning rule derived from information-theoretic and thermodynamic principles. The objective of Boltzmann learning is to adjust the connection weights so that the states of visible units satisfy a particular desired probability distribution. According to the Boltzmann learning rule, the change in the connection weight w_{ij} is given by where η is the learning rate, and $\langle x_i x_j \rangle$ and $\langle x_i x_j \rangle_{\text{clamped}}$ are the correlations between the states of units i and j when the network operates in the clamped mode and free-running mode, respectively.

Learning Algorithm:

- Initialize Weights:

Randomly initialize weights of the network.

- Positive Phase: Present data to the visible units and compute the correlations between visible and hidden units.
- Negative Phase: Run the network freely to generate samples from the model's distribution and compute the correlations again.
- Update Weights: The weights are updated based on the difference between the correlations computed in the positive and negative phases.
- Repeat: Iterate through the training set until convergence.

Challenges of Boltzmann Learning:

- **Computationally Expensive:**

Requires sampling from the model's distribution, which can be very slow (especially for fully connected BMs).

- **Intractable for Large Models:** The unrestricted BM's complexity makes it difficult to scale up.

- **Gibbs Sampling or Contrastive Divergence (CD):** Methods like CD are often used in Restricted Boltzmann Machines to speed up learning.

Applications of Boltzmann Machines:

- **Dimensionality Reduction**
- **Feature Learning**
- **Generative Modeling**
- **Recommendation Systems (e.g., Netflix prize)**
- **Energy-based Modeling**

State transition Diagram and false minima problem

What is State Transition?

- A State can be consider as current condition or situation of system; this can be active, inactive, in
- progress, waiting, stopped and so many states.
- **Transition** is the process or a period of changing state from one condition to another.
- A **State Transition** diagram is a type of diagram used to describe the behavior of systems. State diagrams require that the system described is composed of a finite number of states.

Four Parts of State Transition Diagram

- 1) **States** that the software might get

1st Try

- 2) **Transition** from one state to another state



- 3) **Events** that origin from / to transition

Incorrect Pin

- 4) **Action** can be result from Transition

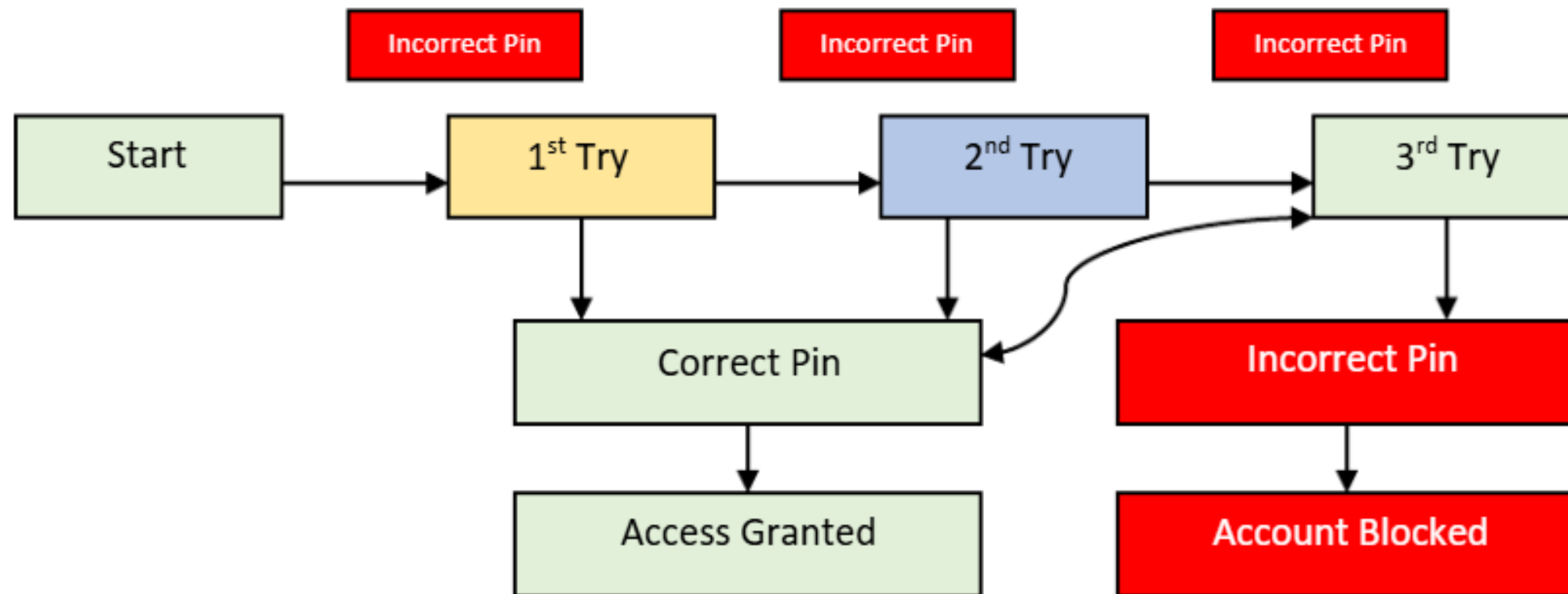
Access Granted

State Transition Table

- In state transition diagram the states are shown in boxed texts, and the transition is represented by arrows. It is also called State Chart or Graph.
- In state transition table all the states are listed on the left side, and the events are described on the top.
- Each cell in the table represents the state of the system after the event has occurred. It is also called State Table.

Example of State Transition Diagram

- Let's consider an ATM system function where if the user enters the invalid password three times the account will be locked.



Conti..

- In this system, if the user enters a valid password in any of the first three attempts the user will be logged in successfully.
- If the user enters the invalid password in the first or second try, the user will be asked to re-enter the password. And finally, if the user enters incorrect password 3rd time, the account will be blocked.
- In the diagram whenever the user enters the correct PIN he is moved to Access granted state, and if he enters the wrong password he is moved to next try and if he does the same for the 3rd time the account blocked state is reached.

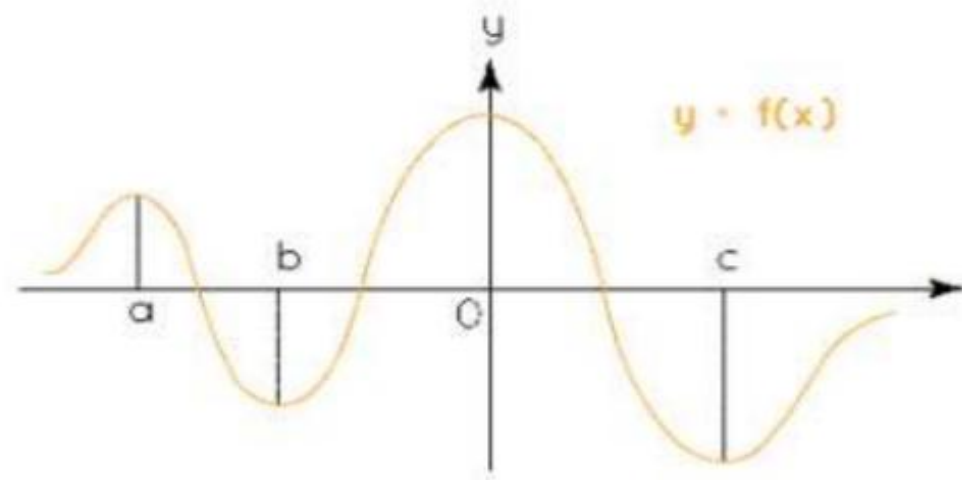
State Transition Table

States	Correct PIN	Incorrect PIN
S1) Start	S5	S2
S2) 1st attempt	S5	S3
S3) 2nd attempt	S5	S4
S4) 3rd attempt	S5	S6
S5) Access Granted	–	–
S6) Account blocked	–	–

False Minima problem-

- Minima is the minimum value of function within the given set of ranges.

Note: Maxima and minima are the peaks and valleys in the curve of a function respectively. Maxima will be the highest point on the curve within the given range and minima would be the lowest point on the curve.



- In the image given below, we can see various peaks and valleys in the graph.
- At $x = a$ and at $x = 0$, we get maximum values of the function,
- and at $x = b$ and $x = c$, we get minimum values of the function.
- All the peaks are the maxima and the valleys are the minima.

What is False Minima Problem?

- False minima, also known as local minima, is a common problem that can occur in optimization
- algorithms used in artificial intelligence. In optimization, the goal is to find the minimum or maximum value of a function.
- However, some functions can have multiple local minima, which are points where the function has a lower value than all of its neighboring points, but are not the absolute minimum of the function.
- The false minima problem occurs when an optimization algorithm gets stuck in one of these local minima instead of finding the global minimum. This can happen when the algorithm follows a path that leads to a local minimum, but is unable to escape and find the global minimum. This can result in suboptimal solutions and reduce the effectiveness of the algorithm.

There are several techniques used in artificial intelligence to address the false minima problem, including:

- **1. Initialization:** The initial values of the parameters can have a significant impact on the optimization process. By selecting initial values that are more likely to lead to the global minimum, the chances of getting stuck in a local minimum can be reduced.
- **2. Exploration vs. exploitation:** Some optimization algorithms balance exploration, which involves searching for new solutions, and exploitation, which involves refining known solutions. By balancing exploration and exploitation, the algorithm can be more effective at finding the global minimum.
- **3. Stochastic optimization:** Stochastic optimization involves introducing randomness into the optimization process. By adding randomness, the algorithm can explore different regions of the search space and reduce the chances of getting stuck in a local minimum.

Stochastic Update

- Stochastic refers to a variable process where the outcome involves some randomness and has some uncertainty.
- Stochastic update is a technique used in machine learning and artificial intelligence to update the parameters of a model based on a random subset of the training data, rather than the entire dataset.
- Stochastic update is a form of stochastic gradient descent, which is a widely used optimization technique in machine learning.
- In stochastic update, the model parameters are updated based on the error or loss computed on a randomly selected subset of the training data, also known as a mini-batch. The mini-batch size is typically much smaller than the full training set, and the random selection of data points helps to introduce randomness and prevent the optimization process from getting stuck in local minima.
- Stochastic update has several advantages, which involves updating the model parameters based on the full training set.

Advantages include:

- 1. Faster convergence: Stochastic update can converge faster than batch update, especially when the dataset is large, as the model parameters are updated more frequently.
- 2. More robustness: Stochastic update can be more robust to noisy or irrelevant data points, as they are less likely to have a significant impact on the model parameters.
- 3. Reduced memory requirements: Stochastic update requires less memory than batch update, as it only needs to store the mini-batch instead of the entire training set.

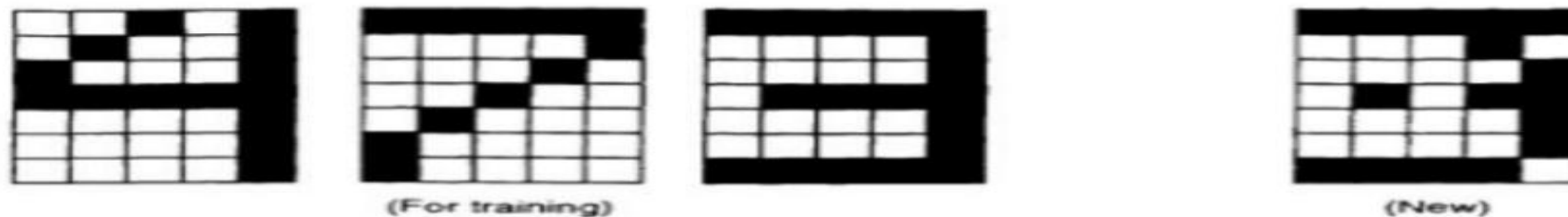
Disadvantages:

- 1. Higher variance: Stochastic update can introduce higher variance in the optimization process, as the model parameters are updated based on a random subset of the training data.
- 2. Slower convergence at the end: At the end of the training process, stochastic update can converge slower than batch update, as the updates become less frequent and more noisy.
- Popular examples of stochastic optimization algorithms are: Simulated Annealing, Genetic Algorithm, Particle Swarm Optimization

Basic functional units of ANN for pattern recognition tasks:

Pattern association-

- Pattern association is the process of memorizing input-output patterns in a hetero associative network architecture, or input patterns only in an auto associative network, in order to recall the patterns when a new input pattern is presented. It is not required that the new input pattern be exactly the same as one that is memorized. It can be different, but similar to it. Three exemplar patterns are shown in figure.
- After memorizing them in a system, a new pattern is presented, a corrupted variant of the pattern 3. An associative memory system should associate this pattern with one of the memorized patterns. This is a task for an auto associative network architecture.



Pattern classification and pattern mapping tasks

- Before searching for a pattern there are some certain steps and the first one is to collect the data from the real world.
- The collected data needs to be filtered and pre-processed so that system can extract the features from the data. This data can be divided into general category like audio, number, image, text which can be in the form of traffic signal images, human faces, handwritten letters, and the DNA sequences. These need to be converted to machine understandable language like 0 and 1.
- In classification, the algorithm assigns labels to data based on the predefined features. This is an example of supervised learning.

What is Pattern Classification?

- Pattern classification is a fundamental problem in artificial intelligence that involves assigning objects or data points to predefined categories or classes. The goal of pattern classification is to develop an algorithm or model that can accurately predict the class of a new data point based on its features or attributes.
- Pattern classification, the algorithm or model is typically trained on a labelled dataset, where each data point is associated with a known class label. The model learns to recognize patterns in the data that are characteristic of each class and uses these patterns to classify new, unseen data points.

There are several techniques used in artificial intelligence for pattern classification, including:

- 1. Supervised learning: This involves training a model on labeled data, where the class labels are known. The model learns to recognize patterns in the data that are characteristic of each class and uses these patterns to classify new, unseen data points.
- 2. Unsupervised learning: This involves training a model on unlabelled data, where the class labels are not known. The model learns to identify patterns in the data without any prior knowledge of the class labels.
- 3. Semi-supervised learning: This involves training a model on a combination of labelled and unlabelled data. The model learns to identify patterns in the labelled data and uses this knowledge to classify the unlabelled data.

Pattern Mapping Task-

- Pattern mapping is a common task in artificial intelligence (AI) and involves identifying and mapping patterns within data. There are several techniques used in AI to perform pattern mapping tasks, including:
 - 1. Machine learning: This involves training an AI model on a large dataset and then using the model to identify patterns in new data.
 - 2. Deep learning: This is a subset of machine learning that uses neural networks to identify patterns in data. Deep learning is particularly useful for tasks that involve image recognition and natural language processing.
 - 3. Clustering: This involves grouping data points based on their similarities. Clustering algorithms can be used to identify patterns in large datasets.

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- 4. Association rule mining: This involves identifying relationships between items in a dataset. Association rule mining is commonly used in market basket analysis to identify products that are frequently purchased together.
- 5. Decision trees: This involves breaking down a dataset into smaller subsets based on specific criteria. Decision trees are commonly used in classification tasks to identify patterns that distinguish between different classes of data.

Overall, pattern mapping tasks in AI involve using algorithms and techniques to identify and extract meaningful patterns from large datasets. These patterns can then be used to make predictions or inform decision-making processes in various domains.

